

Paper RQM-T3 — Anyon Prohibition in 3+1D

A Dimension-Comparison Structural Result

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Preamble: What This Paper Does NOT Claim

1. **The 13 primitives are not derived.** They are postulated per Paper_087.
2. **No claim that anyon prohibition is forced from nothing.** The result is conditional on P06 + P10 + P03 + standard configuration-space topology (Leinaas–Myrheim, Wilczek) + finite-group rep theory.
3. **No claim of numerical-content derivation.** This paper is a pure dimension-comparison structural result; no specific particle-content is inherited within the substantive claim.
4. **No claim that ED is the only consistent substrate ontology.** Other substrate ontologies are conceivable.
5. **No claim that effective 2D surface anyons are substrate-level anyons.** Anyonic statistics observed in fractional-quantum-Hall systems is consistent with the substrate framework as coarse-graining of $D = 3 + 1$ to effective 2D.
6. **No claim of empirical adequacy outside $D = 3 + 1$.** $D = 2 + 1$ admits anyons by braid-group structure.

Abstract

Paper RQM-T3 establishes the prohibition of anyonic statistics in $D = 3 + 1$ as a form-IDENTIFIED structural consequence of P06 (spatial dimension) + P10 (rule-type primitive) + P03 (channel + locus indexing), composed with standard algebraic topology of configuration spaces. The substrate-level mechanism: in $D = 3 + 1$, the configuration-space fundamental group for n identical rule-type carriers is the permutation group S_n (not the braid group B_n); the one-dimensional representations of S_n are exactly the trivial (bosonic, $\eta = +1$) and sign (fermionic, $\eta = -1$) representations. Continuous-phase anyons $\eta = e^{i\theta}$ exist only in braid-group representations, which require $d = 2$ spatial dimensions. The result is form-IDENTIFIED with no inherited numerical particle-content; verdict tier: **M1** per Paper_095 (structural result from primitives + standard math alone). T3 is the dimension-comparison companion to T1 spin-statistics: T1 supplies the polarity-rotation factor, T3 supplies the topological admissibility constraint; together they force the fermion/boson dichotomy exhaustively in $D = 3 + 1$. This paper is in the substrate-ontology research-program lineage of 't Hooft, Wolfram, and the causal-set program — not in the operational-reconstruction lineage of Hardy / CDP / Coecke-Kissinger.

1 Statement of Result

Claim. In $D = 3 + 1$ substrate dimensions (P06), the configuration-space fundamental group for n identical rule-type carriers is the **permutation group** S_n , not the **braid group** B_n . Consequently, only one-dimensional representations of S_n are admissible exchange-phases — namely $\eta = \pm 1$ (the trivial and sign representations) — and anyons are structurally prohibited.

The dichotomy fermion/boson is exhaustive in $D = 3 + 1$. In $D = 2 + 1$, the fundamental group is B_n and continuous-phase anyons exist.

form-IDENTIFIED by P06 + rule-type homotopy classification + standard configuration-space topology. No new postulates.

Verdict: **M1** (pure dimension-comparison / topology-driven structural result; no inherited numerical content).

2 Primitive Inputs

2.1 Primitives invoked

- **P06 (Spatial dimension $D = 3 + 1$)** — the load-bearing primitive: forces S_n rather than B_n .
- **P10 (Rule-type primitive)** — supplies the rule-type carriers whose exchange is considered.
- **P03 (Channel + locus indexing)** — supplies the locus indexing on which configuration space is built.

2.2 Upstream Dependencies

- **I-087:** 13-primitive position paper.
- **I-RQM-T1:** Spin–statistics dichotomy.
- **I-Topology:** Standard algebraic topology of configuration spaces (standard math).
- **I-Braid:** Braid-group representation theory (Leinaas–Myrheim 1977, Wilczek 1982; standard math).

3 Derivation

3.1 Configuration space for n identical rule-types

For n identical rule-type carriers in d spatial dimensions, the configuration space is

$$M_n = (\mathbb{R}^d)^n \setminus \Delta / S_n,$$

where Δ is the diagonal (coincidence set) removed because identical carriers cannot occupy the same locus, and division by S_n identifies configurations related by permutation.

3.2 The fundamental group $\pi_1(M_n)$

Standard configuration-space topology (Leinaas–Myrheim, Wilczek; I-Topology + I-Braid):

- $d = 3$ **spatial (i.e., $D = 3 + 1$):** $\pi_1(M_n) = S_n$ (permutation group).
- $d = 2$ **spatial (i.e., $D = 2 + 1$):** $\pi_1(M_n) = B_n$ (braid group).
- $d = 1$ **spatial:** statistics degenerates (no exchange-path topology).

The dimensional crossover at $d = 3$: in three or more spatial dimensions, any loop in configuration space that effects an exchange can be deformed (without crossing the coincidence diagonal) into a loop that effects the same permutation by a different route. All such loops are homotopically equivalent $\rightarrow$ only the permutation matters $\rightarrow \pi_1 = S_n$.

In two spatial dimensions, this freedom is lost: two carriers in the plane cannot pass through each other, and the order of exchange (clockwise vs counterclockwise) becomes topologically distinct $\rightarrow$ braid structure $\rightarrow \pi_1 = B_n$.

3.3 Representations of S_n vs B_n

S_n representations include:

- Trivial (all permutations $\rightarrow +1$): bosonic.
- Sign / alternating (even permutations $\rightarrow +1$, odd $\rightarrow -1$): fermionic.
- Higher-dimensional representations (Young-diagram irreducibles).

For exchange-phase purposes, only the **one-dimensional** representations of S_n produce a single-valued multiplicative phase factor. There are exactly two: trivial ($\eta = +1$, bosonic) and sign ($\eta = -1$, fermionic).

B_n representations include:

- Trivial (all braids $\rightarrow +1$): bosonic.
- Sign (all generators $\rightarrow -1$): fermionic.
- **Continuous-phase one-dim representations** $\eta = e^{i\theta}$ for $\theta \in [0, 2\pi)$: **anyonic**.
- Higher-dimensional non-abelian representations: non-abelian anyons.

The continuous-phase $\eta = e^{i\theta}$ exists in B_n but not in S_n , because B_n has infinite-order generators (each elementary braid generator has infinite order), while S_n 's transpositions have order 2 (squaring gives identity).

3.4 Substrate-level anyon prohibition in $D = 3 + 1$

In $D = 3 + 1$ (P06), the rule-type carriers' configuration space has $\pi_1 = S_n$. By §3.3, the only admissible one-dimensional exchange-phase representations are $\eta = \pm 1$. **Anyons (continuous-phase $\eta = e^{i\theta}$ with $\theta \notin \{0, \pi\}$) are structurally prohibited.**

This is the substrate-level basis for the empirical fact that in three-dimensional spatial settings, all known particles are either fermions or bosons. The exhaustive dichotomy in $D = 3 + 1$ is FORM-FORCED.

3.5 What is admissible in $D = 2 + 1$

Two-dimensional spatial substrate (hypothetical or effective; e.g., quantum-Hall surface states confined to a plane) has $\pi_1 = B_n$. Anyons are admissible.

Empirically: anyonic statistics has been observed in 2D fractional-quantum-Hall systems (Bar-tolomei et al. 2020; Nakamura et al. 2020). These are **effective 2D systems realized within 3D substrate**; the anyonic statistics applies to the surface degrees of freedom whose effective configuration space is two-dimensional.

The substrate framework is consistent with this: the underlying substrate is $D = 3 + 1$ (no anyons at substrate level), but **effective 2D surface theories** can exhibit anyonic statistics within the surface configuration space. This is a coarse-graining phenomenon, not a substrate-level anyon.

3.6 Connection to spin-statistics (T1)

T1 derived the exchange-phase $\eta = (-1)^{2s}$ for spin- s rule-types in $D = 3 + 1$. Combined with this paper's result that only $\eta = \pm 1$ are admissible:

- $\eta = +1 \Rightarrow 2s \text{ even} \Rightarrow s \in \{0, 1, 2, \dots\}$: bosonic.
- $\eta = -1 \Rightarrow 2s \text{ odd} \Rightarrow s \in \{1/2, 3/2, \dots\}$: fermionic.

No values of s produce admissible anyonic phase. The spin-statistics dichotomy is exactly the two-class limit forced by configuration-space topology.

3.7 Connection to $\text{Cl}(3,1)$ (T2)

T2 derived the four-component spinor structure in $D = 3 + 1$. The four-component Dirac spinor is the algebraic substrate of the half-integer-spin fermionic class. T1 + T2 + T3 together establish: $D = 3 + 1$ supports exactly two statistics classes, with the spinor class algebraically realized by $\text{Cl}(3,1)$.

4 Audit Table

#	Step	Label	Notes
1	P06 $D = 3 + 1$	P	Primitive.
2	P10 rule-type carriers	P	Primitive.
3	Configuration space M_n	I	Standard construction.
4	$\pi_1(M_n) = S_n$ for $d = 3$	I	Standard topology.
5	$\pi_1(M_n) = B_n$ for $d = 2$	I	Standard topology.
6	One-dim S_n reps are trivial + sign	I	Standard rep theory.
7	Continuous-phase B_n reps exist	I	Standard math.
8	Anyons prohibited in $D = 3 + 1$	D-via-I	Step 6 applied to step 4.
9	Anyons admissible in $D = 2 + 1$	D-via-I	Step 7 applied to step 5.
10	Effective 2D surface theories may host anyons	D-via-I	Coarse-graining.
11	T1 + T3 yields fermion/boson exhaustive dichotomy	D-via-I	Step 8 + T1.

5 Falsification Criteria

- **F1: Empirical detection of genuinely 3D anyonic statistics.** A particle exhibiting continuous-phase exchange in a bulk three-dimensional system (not effective 2D surface) refutes the prohibition.
- **F2: Substrate evidence that P06 permits $d = 2$ realization for substrate physics.** Would propagate from F2 of Paper_RQM_T1.
- **F3: Standard topology result $\pi_1(M_n) = S_n$ in $d = 3$ shown to be wrong.** Would refute I-Topology.
- **F4: Empirical detection of anyonic statistics in genuine 3+1D condensed-matter systems (not 2+1D surface states).** Empirical refutation target: observation of continuous-phase $\eta = e^{i\theta}$ exchange in bulk 3D — would refute the dimension-comparison argument.

6 Position Statement

Paper RQM-T3 establishes anyon prohibition in 3+1D as **M1** (form-IDENTIFIED structural dimension-comparison result) under P06 + P10 + P03 with no new postulates. The structural form is form-IDENTIFIED by composition of substrate primitives with standard algebraic topology of configuration spaces and standard finite-group rep theory; no numerical particle-content is inherited because the result is purely about admissible exchange-phase classes. The result joins the Wave-3 Relativistic-QM Bridge family as the dimension-comparison companion to T1 spin-statistics, complementary to T2 ($\text{Cl}(3,1)$ algebraic substrate) and T4 (Dirac equation), serving as the load-bearing topological constraint for the fermion/boson exhaustive dichotomy in $D = 3 + 1$.

This paper does NOT claim that effective 2D anyonic systems are substrate-level anyons — those are coarse-graining phenomena. It does NOT claim numerical content beyond the structural

dichotomy.

This paper is in the Wave-3 Relativistic-QM Bridge series of the Event Density program — a substrate-ontology research program in the lineage of 't Hooft's cellular-automaton interpretation, Wolfram's Ruliad, and the causal-set program (Sorkin et al.); NOT in the operational-reconstruction lineage of Hardy / CDP / Coecke-Kissinger.

Verdict: **M1**.